

# Replacement Engine Control Unit

Complete documentation: the machine, the circuits, the specification, and the research all of it rests on.

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This replaces the **Engine Control Unit** block on page 1 of the Kirloskar troubleshooting wiring diagram, manual 08-GP3-60-001. Every pin, signal and colour in these pages was read out of that document or measured in simulation; where something is inferred rather than confirmed, it says so on the line.

The engine has not been inspected yet. Of twelve items on the unknowns register, two are answered by inference and none is confirmed — every confirmation needs the machine. The CPCB IV+ certification position in Memo 06 is a decision for the machine's owners, not an engineering conclusion. **Read that memo before fitting anything.**

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### 1. Field Handbook

The machine, its interface, the circuits in narrative, and how to run it

### 2. Sensor Reference

Every sensor: what it measures, how we read it, what we do with it

### 3. Circuit Reference

All 17 circuit blocks: schematic, reasoning, simulated result, limits

### 4. Reconciled Specification

Binding source of truth -- pin map, power, deviations, unknowns

### 5. Memo 01 -- Engine Identity

Which engine this actually is, established from five hardware constraints

### 6. Memo 02 -- KG640C Controller

What the existing genset controller does, and what it needs from us

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The 94-pin connector: candidates, sourcing, and what stays unconfirmed

### 8. Memo 04 -- Injector Drive

Bank sharing proved safe, and the drive envelope it has to hit

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Why the S32K148, and the timing argument that actually binds

### 10. Memo 06 -- Standards & Law

ISO, CISPR and the CPCB IV+ certification position

**11. Memo 07 -- BOM & Sourcing**

Costed bill of materials, India sourcing, single-source flags

**12. Memo 08 -- Engine Inspection Brief**

The measurements to take at the machine, grouped by machine state

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How DSE and SEDEMAC build temperature inputs, and what transfers

**14. Phase 1 Research Plan**

The plan the eight memos were written against